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Corporate Social Responsibility and Environmental Management / Volume 33, Issue 1 / pp. 1339-1356

RESEARCH ARTICLE | [Open Access](#) | 

Female Directors and Corporate Environmental Decoupling: The Moderating Influence of Board Characteristics

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First published: 20 October 2025

<https://doi.org/10.1002/csr.70228>[VIEW METRICS](#)Accessibility issue? [Request accessibility update.](#)

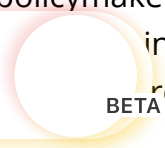
[Correction added on 23 October 2025, after first online publication: In the author byline and Correspondence section, a hyphen has been added to separate the authors' last names in this version.]



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ABSTRACT

This study explores how the relationship between board gender diversity and corporate environmental decoupling is moderated by board characteristics. Based on US listed companies in environmentally non-sensitive industries for the period 2011–2021, our results highlight that female directors tend to mitigate environmental decoupling. However, this effect is strengthened in boards with longer tenure and greater ESG experience but remains insignificant in boards with higher independence. Our findings highlight the interconnectedness of board characteristics, showing that analyzing board gender diversity in isolation may yield biased conclusions. These findings will help policymakers and professional managers in developing effective strategies to reduce environmental decoupling based on board ESG experience.

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The growing pressure on firms to meet sustainability expectations has led to the proliferation of ESG strategies that widen the gap between ESG reporting and ESG performance, a phenomenon known as “ESG decoupling” (Ruiz-Blanco et al. 2022; Lassoued and Khanchel 2023). This disconnect is often perceived as unethical corporate behavior, attracting widespread criticism regarding the role of boards of directors in overseeing ESG practices (Deegan 2017; Eliwa et al. 2023; Talpur et al. 2024).

In recent years, research has increasingly emphasized the significance of board gender diversity in shaping ESG strategies. Nonetheless, empirical evidence on the influence of female directors on ESG decoupling remains limited and inconclusive (Shahab et al. 2022; Eliwa et al. 2023; Gull, Hussain, Khan, Khan, and Saeed 2023). Some emerging studies suggest that these contradictory findings may stem from contextual factors moderating the effect of board gender diversity, such as religiosity (Eliwa et al. 2023) and environmental uncertainty (Qureshi et al. 2024). However, the moderating effect of board characteristics in the role of female directors remains unexplored.

Our study focuses on corporate environmental decoupling, a fundamental pillar of ESG strategies, and aligns with this prior literature arguing that a one-size-fits-all approach fails to capture the nuanced role of female directors in managing this decoupling. Specifically, we seek to fill in a critical research gap by examining how board characteristics serve as key moderating factors in the relationship between female directors and corporate environmental decoupling in environmentally non-sensitive industries. In particular, we aim to determine whether the effect of female directors depends on three board attributes: ESG experience, independence, and tenure. This constitutes a novel research question and is both timely and relevant for several reasons.

First, environmental decoupling has become a major concern for policymakers (European Commission 2022; ESMA 2023; IOSCO 2023) and professional organizations (Deloitte 2022; Ernst and Young 2022; PwC 2022) due to its detrimental societal consequences. This decoupling undermines the credibility and trustworthiness of corporate reports (Boiral et al. 2019), leading to reduced corporate transparency and greater information asymmetry, and creating inefficiencies in capital markets (Di and Li 2023). Furthermore, environmental decoupling can harm firms by damaging corporate reputation (Miras-Rodríguez et al. 2020), restricting access to finance (García-Sánchez et al. 2021), and reducing market valuation (Hawn and Ioannou 2016).

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Second, board gender diversity influences legislative frameworks and corporate governance codes, and is widely regarded in the

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focused on the impact of board gender diversity on either ESG reporting or ESG performance in isolation, thus limiting our understanding of its broader influence on environmental strategies. Consequently, it is crucial to examine the simultaneous effects of female directors on both environmental reporting and environmental performance to develop a more comprehensive perspective on their role in shaping corporate environmental outcomes (Eliwa et al. 2023; Gull, Hussain, Khan, Khan, and Saeed 2023). Additionally, exploring the interplay between board gender diversity and other board characteristics remains a critical topic in corporate governance, essential for understanding the actual influence of female directors (Bel-Oms et al. 2024; Chang et al. 2024; Gavana et al. 2024).

Third, both practitioners and policymakers are increasingly focused on understanding industry-specific ESG reporting (Khamisu et al. 2024). As a result, various frameworks, such as the Global Reporting Initiative (GRI), the European Sustainability Reporting Standards under the Corporate Sustainability Reporting Directive (EU 2022/2464), and the Sustainability Accounting Standards Board (SASB), advocate for industry-specific sustainability guidelines. Recent studies underscore the importance of considering industry environmental sensitivity when analyzing the relationship between boards and environmental outcomes (Qureshi et al. 2020; Zaiane and Ellouze 2023). Therefore, investigating the role of board gender diversity in mitigating environmental decoupling in environmentally non-sensitive industries is essential for developing policies and corporate strategies that foster genuine ESG commitment across sectors.

Based on a sample of U.S. firms from 2011 to 2021, our findings highlight that the effect of board gender diversity in reducing corporate environmental decoupling is contingent on certain board characteristics. While this relationship is not moderated by board independence, the influence of female directors on corporate environmental decoupling is stronger in boards with longer tenure and greater ESG experience. These results contribute to the literature in several ways.

First, we contribute to the existing empirical evidence regarding the influence of corporate governance mechanisms on ESG decoupling, with a specific focus on its environmental pillar, which is a critical dimension. Previous studies have explored how ESG decoupling can be influenced by external factors, such as the institutional environment (Tashman et al. 2019), regulations (Aboud et al. 2024), or analysts (Zhang et al. 2022), as well as by organizational characteristics (Kim and Lyon 2015), and board characteristics (e.g., board independence, board network; board committees, CEOs) in ESG decoupling (Sauerwald and Su 2019; Yu et al. 2020; Gull, Hussain, Khan, Mushtaq, and

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results provide novel insights into the dynamics between specific board characteristics, highlighting the interconnectedness of directors and suggesting that analyzing the isolated effects of individual director characteristics may lead to biased conclusions in research. In this regard, we explain the interactions among several relevant board characteristics (gender diversity, tenure, ESG experience, and independence), which are of significant interest to policymakers, professionals, and academics alike.

Second, more specifically, our evidence contributes to the limited literature on board gender diversity and corporate environmental decoupling. To date, empirical findings have been inconclusive (Shahab et al. 2022; Eliwa et al. 2023; Gull, Hussain, Khan, Khan, and Saeed 2023; Gull et al. 2024), likely because the impact of female directors on ESG decoupling varies depending on different contextual factors. Our findings support the notion that a more nuanced approach is required to fully understand the effects of female directors on ESG strategies. In this context, our study provides a pioneering perspective, confirming that the relationship between female directors and corporate environmental decoupling is contingent upon specific characteristics of the board of directors. This evidence underscores the importance of moving beyond a one-size-fits-all approach in research on the effects of female directors.

Third, our analysis of how board gender diversity affects corporate environmental decoupling in environmentally non-sensitive industries makes a significant contribution to this research field. The environmental sensitivity of an industry is a crucial factor in understanding the relationship between boards and ESG outcomes (Qureshi et al. 2020; Zaiane and Ellouze 2023; Appiagyei and Donkor 2024). Environmentally non-sensitive industries typically face fewer regulations and experience less pressure from stakeholders to adopt environmentally responsible practices (Miralles-Quirós et al. 2018; Barroso-Méndez et al. 2024). Therefore, our findings offer valuable insights into the role of board gender diversity in managing corporate environmental decoupling in industries that may be more susceptible to engaging in unethical behaviors related to environmental practices.

The rest of the paper is organized as follows. Section 2 provides a literature review and outlines the hypothesis development. Section 3 details the research methodology, while Section 4 presents the empirical analyses. Lastly, Sections 5 and 6 cover the discussion and conclusions, respectively.

2 Theoret

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sustainability decoupling in general, considering ESG as a whole (Eliwa et al. 2023; Gull, Hussain, Khan, Nadeem, and Zalata 2023), others find that the association between board gender diversity and ESG decoupling may be negative (Shahab et al. 2022) or non-significant (Gull, Hussain, Khan, Khan, and Saeed 2023). In terms of specific environmental decoupling, research is still very incipient and further empirical evidence is needed (Gull et al. 2024). Theoretical perspectives can provide insights into the potential mechanisms through which female board representation influences corporate environmental decoupling. Although no single theory fully explains this relationship, stakeholder theory and gender socialization theory act as complementary theories offering valuable frameworks for understanding the role of female directors in corporate environmental strategies, providing a comprehensive explanation (Vu and Dang 2021; Heubeck 2024).

On the one hand, from a stakeholder theory perspective, boards of directors are key to aligning corporate strategies with stakeholder interests, particularly regarding environmental sustainability (Harjoto et al. 2015). A diverse board, especially one with greater female representation, enhances firms' responsiveness to sustainability demands by fostering long-term value creation (Wood and Eagly 2009) and improving ESG awareness (Arayssi et al. 2016; Wasiuzzaman and Subramaniam 2023). Their presence also strengthens stakeholder-oriented governance, reducing the likelihood of symbolic environmental commitments and promoting substantive sustainability policies (Isidro and Sobral 2015). Ultimately, gender-diverse boards can drive firms toward more transparent and responsible environmental strategies, ensuring alignment with both regulatory expectations and long-term corporate resilience (Manita et al. 2018). Consequently, female directors' stronger stakeholder orientation helps align corporate environmental policies with actual reporting, thereby reducing environmental decoupling.

On the other hand, gender socialization theory highlights behavioral differences between male and female directors, attributing them to distinct psychological traits. These variations suggest that women in leadership roles can significantly influence corporate environmental strategies. Indeed, this theory can offer an explanation for what manner and reasons female directors react to environmental pressures. Research indicates that female directors often exhibit personal traits, such as empathy and supportiveness (Eagly et al. 2003) and a focus on long-term well-being (Alkhawaja et al. 2023), which make them particularly sensitive to sustainability issues and attuned to stakeholder concerns (Arayssi et al. 2016; Harjoto and Rossi 2019). Have questions about this content? **Try AI Companion.** Enhance awareness of corporate environmental strategies and encourage a more proactive approach to addressing sustainability challenges at the strategic level (Atif and

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decoupling.

Furthermore, the effect of board gender diversity on corporate environmental decoupling may be particularly relevant in environmentally non-sensitive industries, where firms might be less concerned about environmental practices (Hamza and Jarboui 2021), and therefore the board of directors is likely to be a critical factor in shaping environmental decisions (Qureshi et al. 2020; Zaiane and Ellouze 2023). Firms operating in environmentally non-sensitive industries often face fewer regulatory constraints and less pressure from stakeholders to adopt sustainable practices (Barroso-Méndez et al. 2024; Miralles-Quirós et al. 2018). In this context, female directors may act as an internal governance mechanism that substitutes for weak external pressures (Aljanadi et al., Aljanadi 2025), and drawing on stakeholder theory and gender socialization theory, they can exert a significant influence on environmental strategies and mitigating environmental decoupling. According to the previous argument, the following hypothesis is formulated:

H1. *Board gender diversity mitigates corporate environmental decoupling in environmentally non-sensitive industries.*

Given that directors operate within a specific environment that shapes their decision-making, the influence of female board members on corporate environmental decisions is likely to depend on contextual factors (Bravo and Reguera-Alvarado 2019; Endrikat et al. 2021). In the field of ESG decoupling, the literature recognizes that the impact of female directors on this phenomenon may be contingent on factors such as religiosity (Eliwa et al. 2023) and economic policy uncertainty (Qureshi et al. 2024). However, board dynamics, shaped by its characteristics, may act as a key moderating factor in the relationship between board gender diversity and corporate environmental decoupling, as they affect the board's monitoring and oversight capabilities. Building on agency theory, we examine three potential moderating factors: board independence, tenure, and ESG experience. The following paragraphs provide theoretical arguments for each factor, explaining how they may influence board orientations toward environmental decision-making and, in particular, the effect of female directors on corporate environmental decoupling.

2.1 Board Independence

From an agency theory perspective, female directors are expected to act as impartial overseers, ensuring that firms integrate environmental considerations into their strategies with sta-

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and Jiang 2019), leading to more robust environmental commitments and a reduction in symbolic environmental actions that contribute to decoupling (Taglialatela et al. 2024). Moreover, independent directors tend to have broader connections with external stakeholders and institutional actors, and their outsider perspective enables them to exert pressure on management to adopt meaningful ESG policies rather than merely symbolic gestures (Nadeem et al. 2020).

Nevertheless, board independence might also increase corporate environmental decoupling. In this regard, independent directors may lack deep firm-specific knowledge regarding sustainability initiatives, which, based on agency theory, could limit their ability to effectively monitor environmental strategies (Guerrero-Villegas et al. 2018). Another potential drawback is that independent directors, despite their theoretical autonomy, can be influenced by large shareholders who choose financial returns over sustainability concerns (Gallego-Álvarez and Pucheta-Martínez 2020), reducing their ability to control and resulting in a weakened commitment to environmental initiatives and a higher degree of decoupling. Additionally, excessive board independence may lead to inefficiencies and conflicts within the boardroom by reducing cohesion among board members, hindering collaboration, and lowering the oversight of environmental decision-making processes (Guldiken and Darendeli 2016). To sum up, board independence may lead to either increased or decreased board monitoring. Our second hypothesis considers that the effect of female directors on environmental decoupling may depend on how women act under different levels of board monitoring driven by board independence. On the one hand, contexts of heightened board monitoring may either enhance or limit the differential influence of female directors. When monitoring is effective, female directors can more actively impact environmental strategies and couple reporting with performance. However, if monitoring is already high or rigid, their unique influence may be less visible. Conversely, under conditions of low board monitoring, female directors may either compensate for limited board oversight and increase alignment between reporting and practices, or face minimal influence if independence and monitoring mechanisms are insufficient.

Therefore, the following hypothesis is formulated:

H2. *The effect of board gender diversity on corporate environmental decoupling is moderated by board independence in environmentally non-sensitive industries*

2.2 Board Te

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capabilities. Directors with extensive tenure accumulate valuable expertise regarding the firm's internal operations, external environment, and strategic decision-making processes (Bravo and Reguera-Alvarado 2017), which enables more effective oversight of corporate strategies, including sustainability-related initiatives (Chen 2013). Specifically, these resources are likely to enable longer-tenured directors to better align environmental commitments with business operations, promoting a more integrated and authentic approach to corporate sustainability (Patro et al. 2018). Their tenure also provides a historical perspective on the firm's ESG-related efforts, ensuring a more complete oversight of environmental reporting (Harjoto et al. 2015). Furthermore, directors with longer tenures often possess extensive networks and stronger ties to industry and regulatory bodies, which may foster greater environmental accountability (Vafeas 2003; Kor and Sundaramurthy 2009).

However, excessive board tenure may exacerbate corporate environmental decoupling.

Based on agency theory, directors who have served for extended periods may develop close social bonds with management, reducing their monitoring role and willingness to critically evaluate corporate environmental strategies (Vafeas 2003; Dalton and Dalton 2007). This diminished oversight can result in symbolic environmental actions that give the illusion of corporate responsibility without genuine commitment (Al-Qahtani and Elgharbawy 2020). Over time, entrenched board members may resist change, leading to outdated environmental policies and ineffective governance (Khan et al. 2021). Additionally, prolonged tenure can foster insular decision-making, where directors prioritize internal consensus over external accountability, limiting the board's responsiveness to evolving sustainability concerns (Ben-Amar et al. 2013). This may also reduce engagement with external stakeholders, hindering the board's ability to adapt to changing societal expectations regarding sustainability (Kim and Yang 2014).

Therefore, board tenure may lead to either increased or decreased board monitoring. Our third hypothesis considers that the effect of female directors on environmental decoupling may depend on how women act under different levels of board monitoring associated with tenure. When tenure increases board monitoring, female directors can either have more room to actively mitigate environmental decoupling or see their influence become less visible. Conversely, when tenure reduces board monitoring, female directors may compensate for limited oversight and mitigate environmental decoupling, or their impact may be minimal if monitoring mechanisms are weak.

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Therefore, the following hypothesis is formulated.

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2.3 Board ESG Experience

The experience of board members in ESG issues can play a crucial role in mitigating corporate environmental decoupling by enhancing the board's ability to integrate sustainability into decision-making processes. Boards with ESG experience can serve an essential monitoring function since directors with ESG knowledge can improve oversight of stakeholder-oriented perspectives, fostering long-term sustainability (Shepherd et al. 2017). Furthermore, according to agency theory, ESG experience equips boards with the ability to develop and oversee substantive environmental actions, as this experience provides firms with valuable external resources, including knowledge, professional skills, and social networks in sustainable development areas. Thus, ESG experience on boards encourages firms to adopt more meaningful and actionable sustainability strategies rather than engaging in superficial compliance efforts (Asad et al. 2023). During ESG decision-making processes, board members with this experience enhance the comprehension of ESG values, improve the forecasting of ESG trends, and facilitate the identification of related opportunities and challenges (Wang et al. 2025). This leads to ESG decisions that are more informed (Adomaityte-Subaciene et al. 2020; Man Zhang and Greve 2019) and enhances firms' strategic positioning in the face of increasing external pressures and shifting market expectations toward sustainable practices (Durand et al. 2019). In addition, directors with ESG experience are also more inclined to foster a corporate culture that puts sustainability first, thereby embedding ESG considerations into organizational values and decision-making frameworks (Menicucci and Paolucci 2023).

Accordingly, greater board ESG experience would lead to increased board monitoring. Our fourth hypothesis considers that, in contexts of heightened board monitoring due to greater board ESG experience, female directors may either have more scope to actively mitigate environmental decoupling or see their influence less visible.

Therefore, the following hypothesis is formulated:

H4. *The effect of board gender diversity on corporate environmental decoupling is moderated by board ESG experience in environmentally non-sensitive industries.*

3 Research Method

3.1 Data and

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We analyze firms which belong to environmentally non-sensitive industries listed on S&P 500 for the period from 2011 to 2021, building a final sample of 1645 firm-year observations, due to

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corporate environmental decoupling in firms that tend to face fewer regulations and experience less pressure concerning environmental issues.

The data for the design of our variables was obtained from broadly recognized databases. First, we extracted corporate environmental decoupling items, referring to both environmental performance and environmental reporting, from the LSEG Eikon database. Second, financial data were collected from Compustat. Finally, information concerning female representation on boards as well as other board characteristics was obtained from BoardEx.

3.2 Variables Design

3.2.1 Dependent Variable: Corporate Environmental Decoupling

Following prior studies (Hawn and Ioannou [2016](#); Tashman et al. [2019](#); García-Sánchez et al. [2022](#); Eliwa et al. [2023](#); Aboud et al. [2024](#); Gull et al. [2024](#)), we define the variable corporate environmental decoupling (*ENV_DEC*) as the difference between the current environmental reporting (external actions) and the lagged environmental performance (internal actions). First, we considered 22 items related to the disclosure of environmental information (reporting) and 21 items connected with environmental initiatives (performance) obtained from LSEG Eikon. Second, we created two intermediate variables: environmental reporting and environmental performance as the sum of all the scores of each variable. Third, we calculated *ENV_DEC* as the difference between the previous intermediate variables and we normalized it considering the sample values.

3.2.2 Explanatory Variables and Control Variables

Our main explanatory variable is board gender diversity. In line with previous studies (Francoeur et al. [2019](#); Aboud et al. [2024](#); Eliwa et al. [2023](#)), we employ the percentage of female board members (*Prop_Female*). Furthermore, in order to ensure that the findings of our empirical models are not biased by the design of our variable, we also consider an alternative proxy of board gender diversity, which relies on the Blau index of heterogeneity (*Blau*) (Blau [1977](#)).

In addition, our study incorporates both board characteristics and firm-level characteristics that may also affect corporate environmental decoupling. In relation to the board of directors, we control the effect of independent board members, of independent directors within the board; board members' ESG experience (*ESG_Exp*), defined as the proportion of directors who are also on other boards from firms that operate in environmentally sensitive industries;

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directors belong; board size (*Size*), rated as the total number of directors on the board; and the existence of a sustainability or CSR committee (*CSR_com*), a dichotomous variable coded as 1 in the presence of this committee and 0 otherwise. As regards firm-specific variables, we included firm size (*FSize*), defined as the natural logarithm of total assets; firm performance (*FPerf*), computed as the return on assets; and firm leverage (*FLev*), calculated as the ratio of total debt to total assets. Finally, we include a set of year dummies.

Table 1 presents all the variables' definitions.

TABLE 1. Variables description.

Name	Variable definition
ENV_DEC	Difference between environmental reporting and environmental p
Prop_Female	Percentage of female board members
Blau	Blau index of heterogeneity
Indep	Proportion of independent board members
Tenure	Average length of tenure of board members
ESG_Exp	Proportion of directors with specific experience in other boards from firms that operate in
Age	Standard deviation of the age of board members
Directorships	Average number of other boards to which directors belo
Size	Total number of directors on the board
CSR_com	Dummy variable that equals 1 if the company represents this committe
FSize	Natural logarithm of total assets
FPerf	Return on assets
FLev	Ratio total debt to total assets

3.3 Empirical model

Our general model of corporate environmental decoupling, which is reflected in Hypothesis H1. This model is presented in

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$$\begin{aligned} \text{ENV_DEC}_{it+1} = & \beta_0 + \beta_1 \text{Prop_female (Blau)}_{it} + \beta_2 \text{Indep}_{it} \\ & + \beta_3 \text{Tenure}_{it} + \beta_4 \text{ESG_Exp}_{it} + \beta_5 \text{Age}_{it} \\ & + \beta_6 \text{Directorships}_{it} + \beta_7 \text{Size}_{it} + \beta_8 \text{CSR_com}_{it} \\ & + \beta_9 \text{FSize} + \beta_{10} \text{FPerf}_i + \beta_{11} \text{FLev}_t + \beta_{11} \text{Year}_t + \varepsilon_{it} \end{aligned}$$

(1)

Additionally, to test the hypotheses [H2–H4](#), we performed moderation analyses to test how board of directors' characteristics can affect the influence of female directors on corporate environmental decoupling. To that end, interaction variables are included between the measures for board gender diversity and each board characteristic.

In order to enhance the robustness of our results, endogeneity has been considered in all the models, which have been estimated by using both a generalized method of moments (GMM) and fixed effects. On the one hand, GMM (Arellano and Bond [1991](#)) allows addressing simultaneity and omitted variable bias (Shahab et al. [2022](#)). To ensure the validity of our instruments, the results of the Hansen test and Sargan test of over-identifying restrictions are reported in the results tables. On the other hand, the use of a fixed effect with the lagged main independent variable (García-Sánchez et al. [2022](#)) enables considering the possibility that current values of gender diversity on boards might be endogenous to our dependent variable.

In addition, all the models have been estimated by alternatively considering our two measures for board gender diversity: *Prop_Female* and *Blau*.

4 Results

4.1 Descriptive Statistics and Correlations

Table [2](#) shows the main descriptive statistics. The average score of *ENV_DEC* is -0.089 , which is in line with other recent studies (Gull, Hussain, Khan, Nadeem, and Zalata [2023](#); Gull, Hussain, Khan, Khan, and Saeed [2023](#); Aboud et al. [2024](#); Gull et al. [2024](#)). Concerning the main explanatory variables, the proportion of women on boards (*Prop_Female*) is 23% and the Blau index (*Blau*) is 0.34. Regarding the moderating variables, 85% of directors are independent. Board average tenure is about 10 years, ² and 43% of directors have ESG experience.

TABLE 2. Descrip

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ENV_DEC	1882	−0.089	0.163	−0.196	−0.078	0.012
Prop_female	1882	0.233	0.105	0.167	0.231	0.3
Blau	1882	0.335	0.110	0.278	0.355	0.42
Indep	1882	0.85	0.084	0.818	0.889	0.909
Tenure	1882	10.886	5.568	7.041	9.617	13.664
ESG_Exp	1882	0.434	0.450	0	0.167	1
Age	1882	6.879	2.046	5.5	6.6	8
Directorships	1882	0.217	0.173	0.090	0.182	0.333
Size	1882	10.762	2.211	9	11	12
CSR_com	1882	0.903	0.300	1	1	1
FSize	1882	16.639	1.182	15.834	16.625	17.332
FPerf	1882	0.082	0.075	0.039	0.064	0.112
FLev	1882	3.244	4.100	1.097	1.960	3.618

Table 3 reports the correlation matrix. In line with our expectations, the results show a significant negative correlation between board gender diversity and corporate environmental decoupling. Given that a lack of multicollinearity is broadly accepted when VIF values are below 5 (Studenmund 1997; Abid et al. 2021; Farooq et al. 2022), we deem that multicollinearity is not an issue in our sample.

TABLE 3. Correlation matrix.

Variables	(2)	(3)	(4)	(5)	(6)	(7)
(1) ENV_DEC	−0.097***	−0.090***	0.003	0.038*	−0.204***	−0.015
(2) Prop_female		0.233***	0.105***	0.167***	0.231	−0.085***
(3) Blau			0.110***	0.278***	0.355	−0.090***
(4) Indep				−0.126***	−0.054**	−0.264***

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(5)	Tenure					0.039*	−0.023
(6)	ESG_Exp						−0.003
(7)	Age						
(8)	Directorships						
(9)	Size						
(10)	CSR_com						
(11)	FSize						
(12)	FPerf						
(13)	FLev						
	WFE	1.11	1.11	1.22	1.24	1.10	1.12

Note: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

4.2 Regression Results

The main objective is to address the impact of gender diversity on corporate environmental decoupling. In this section, we present the results of all the regression analyses performed. The main results are displayed in Table 4. Our findings demonstrate that female directors exert a significant negative impact on corporate environmental decoupling, which means that women in the boardroom try to align both environmental reporting and initiatives. These results are confirmed for both measures of board gender diversity (*Prop_female* and *Blau*), and for both GMM and fixed effect models. Hence, H1 is supported.

TABLE 4. The relationship between board gender diversity and corporate environmental decoupling.

Variables	GMM	GMM	Fixed effect	Fixed effect
	<i>ENV_DEC</i>	<i>ENV_DEC</i>	<i>ENV_DEC</i>	<i>ENV_DEC</i>
Prop_female				

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	ENV_DEC	ENV_DEC	ENV_DEC	ENV_DEC
Lag Prop_female			−0.127*** (0.047)	
Blau		−0.229** (0.103)		
Lag Blau				−0.133*** (0.041)
Indep	0.082 (0.076)	0.358* (0.197)	0.112* (0.067)	0.118* (0.067)
Tenure	0.000	0.001	0.001	0.001

Note: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

In the next phase of our empirical analysis, the moderating effect of board characteristics on the relationship between female directors and corporate environmental decoupling is tested. First, Table 5 fails to find a significant moderating impact of directors' independence on the association between board gender diversity and corporate environmental decoupling. Thus, Hypothesis H2 cannot be supported.

TABLE 5. Moderating effect of board independence.

Variables	GMM	GMM	Fixed effect	Fixed effect
	ENV_DEC	ENV_DEC	ENV_DEC	ENV_DEC
Prop_female*Indep	5.000 (4.780)			
Prop_female				
Blau*Indep		4.568		

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	ENV_DEC	ENV_DEC	ENV_DEC	ENV_DEC
		(3.777)		
Blau		−3.846		
		(3.183)		
Lag Prop_female*Tenure			0.435	
			(0.448)	
Lag Prop_female			−0.494	
			(0.386)	

Note: Standard errors in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

Second, Table 6 displays a significant negative moderating effect of board tenure on the influence of female board members on corporate environmental decoupling. In this vein, female directors with higher tenure contribute to aligning disclosure and environmental initiatives on boards. These results are reported in all the models. Hence, Hypothesis H3 can be accepted. Figure 1 also corroborates these findings.

TABLE 6. Moderating effect of board tenure.

Variables	GMM	GMM	Fixed effect	Fixed effect
	ENV_DEC	ENV_DEC	ENV_DEC	ENV_DEC
Prop_female*Tenure	−0.143 [*]			
	(0.076)			
Prop_female				
Blau*Tenure		−0.093 [*]		

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	ENV_DEC	ENV_DEC	ENV_DEC	ENV_DEC
		(0.053)		
Blau		1.064 [*]		
		(0.586)		
Lag Prop_female*Tenure			−0.011 [*]	
			(0.007)	
Lag Prop_female			−0.017	
			(0.080)	

Note: Standard errors in parentheses.

- *** $p < 0.01$.
- ** $p < 0.05$.
- * $p < 0.1$.

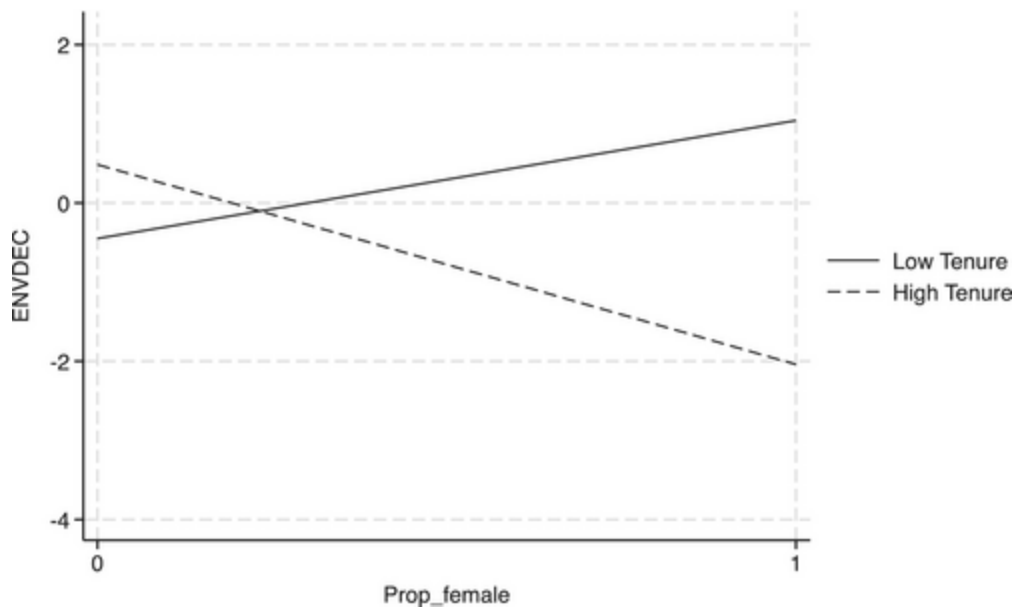


FIGURE 1

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Graphical representation of the moderating effect of board tenure.

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is shown to be negative and significant, highlighting that women tend to decrease corporate environmental decoupling when boards of directors possess greater ESG experience. These findings are confirmed regardless of the use of GMM and the fixed effect, as well as for our two measures of board gender diversity. Therefore, we accept our Hypothesis H4. This result is also illustrated in Figure 2.

TABLE 7. Moderating effect of board ESG experience.

Variables	GMM	GMM	Fixed effect	Fixed effect
	0.028	0.014	0.018	0.017
	(0.026)	(0.023)	(0.024)	(0.024)
FSize	−0.020	−0.037	0.002	0.003
	(0.026)	(0.025)	(0.012)	(0.012)
FPerf	−0.695	−0.834	−0.126 ^{**}	−0.116 [*]
	(0.502)	(0.538)	(0.064)	(0.064)
FLev	−0.006 [*]	−0.002	0.001	0.001
	(0.003)	(0.002)	(0.001)	(0.001)
Year dummies	Yes	Yes	Yes	Yes
Observations	1645	1645	1645	1645
Wald chi2	381.19 ^{***}	592.74 ^{***}		
Sargan test	9.52	9.59		
Hansen test	−0.12	0.96		
R-squared			0.050	0.054
F-test			4.13 ^{***}	4.49 ^{***}

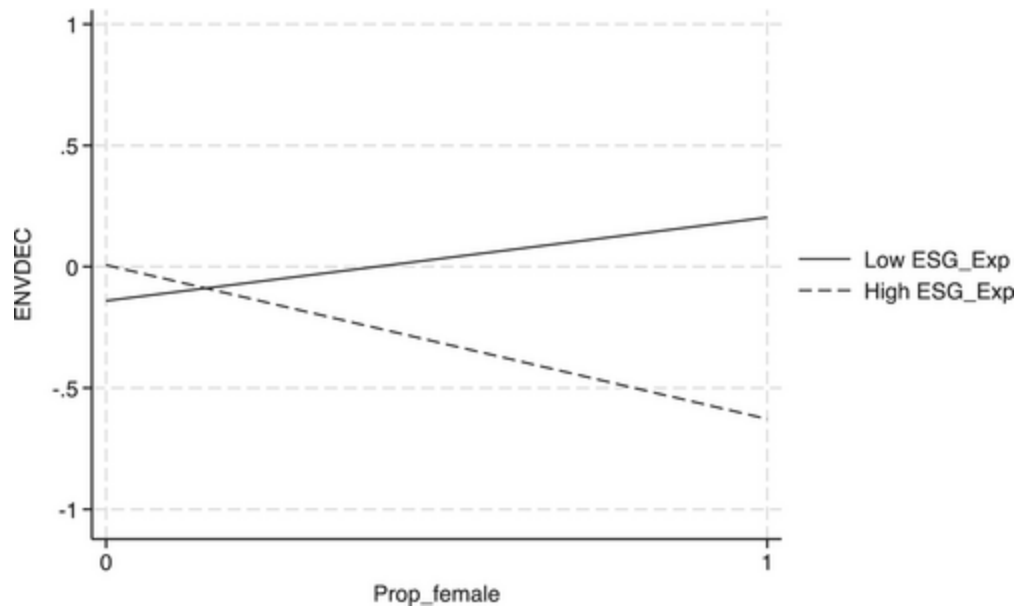
Note: Standard error

*** $p < 0.01$.

** $p < 0.05$.

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[< Back](#)**FIGURE 2**
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Graphical representation of the moderating effect of board ESG experience.

5 Discussion

Our findings underline how the effect of board gender diversity on corporate environmental decoupling in environmentally non-sensitive industries is moderated by board attributes and provides new insights in this research area. Specifically, our paper extends previous research in several ways.

First, our findings contribute to the existing literature on the determinants of environmental decoupling by underscoring the pivotal role of corporate boards. Prior research has identified both external and organizational factors as key drivers of ESG decoupling, with organizational factors encompassing firm-level characteristics and corporate governance mechanisms (Velte 2023; Cepeda et al., Cepêda et al. 2025). Our study advances this discourse by emphasizing the significance of corporate governance, particularly board characteristics, in shaping the market pressure from stakeholders in environmentally non-sensitive industries.

Second, our findings suggest that firms in environmentally non-sensitive industries, which face less market pressure from stakeholders and therefore may be more susceptible to engaging in unethical environmental practices (Miralles-Quirós et al. 2018).

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stakeholder pressures (Velte 2023). With regard to boards of directors, prior studies have examined the influence of CEOs (Sauerwald and Su 2019; Shahab et al. 2022; Gull, Hussain, Khan, Mushtaq, and Orij 2023), governance committees (Gull, Hussain, Khan, Khan, and Saeed 2023), sustainability committees (Gull et al. 2024), and other board characteristics such as directors' networks (Zhao et al. 2022) and director independence (Yu et al. 2020).

Notwithstanding, these variables have been studied individually, and there is no prior evidence on the effect of board characteristics interaction in the context of corporate environmental decoupling. Our findings contribute to this discussion by demonstrating that the board of directors serves as a crucial mechanism for mitigating such decoupling. Specifically, our results offer novel insights into the complex interplay of board characteristics and underscore the interconnected nature of directors' attributes. This perspective suggests that analyzing individual director characteristics in isolation may lead to incomplete or misleading conclusions. Accordingly, we highlight the interdependencies among board gender diversity, independence, tenure, and ESG experience, stressing the importance of considering their interactions when addressing corporate environmental decoupling.

Third, our findings contribute more specifically to the literature on the relationship between board gender diversity and corporate environmental decoupling. The limited existing evidence on this topic remains inconclusive, with prior research suggesting that the presence of female directors may have positive, negative, or non-significant effects on ESG decoupling—both in general and specifically regarding environmental decoupling (Eliwa et al. 2023; Gull, Hussain, Khan, Nadeem, and Zalata 2023; Shahab et al. 2022; Gull et al. 2024). Our study helps reconcile these conflicting findings by supporting the argument that contextual approaches are crucial for a more nuanced understanding of the influence of female board members on environmental decision-making (Bravo and Reguera-Alvarado 2019; Endrikat et al. 2021). On the one hand, in line with hypothesis H1 and consistent with stakeholder theory and gender socialization theory, our results suggest that female directors' stronger stakeholder orientation, combined with their personal traits, enables them to exert a significant influence on environmental strategies and reduce environmental decoupling. On the other hand, our results highlight that board-level characteristics play a key moderating role in the relationship between board gender diversity and corporate environmental decoupling, underscoring the importance of considering the broader governance context in which women operate to fully comprehend the actual influence of female directors. In this regard, our findings indicate that, under conditions of high stakeholder orientation, female directors play a significant role in environmental decoupling.

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demonstrate that both factors moderate the relationship between board gender diversity and corporate environmental decoupling.

Regarding board tenure, longer tenure enhances monitoring of environmental decisions by providing directors with firm-specific knowledge, oversight capabilities, and the ability to align environmental commitments with business operations (Bravo and Reguera-Alvarado 2017; Chen 2013; Patro et al. 2018). This, in turn, amplifies the impact of female directors on corporate environmental decoupling, consistent with the arguments from our hypothesis H3. In environmentally non-sensitive industries, such experience is crucial for women on boards to translate their stakeholder orientation and personal values into concrete sustainability actions. Directors with longer tenure bring deep knowledge, internal networks, and credibility, which strengthens women's ability to align environmental reporting with actual practices. Thus, effective board monitoring, driven by tenure, enhances the ability of female directors to influence environmental strategies, reinforcing their impact on reducing environmental decoupling.

Similarly, board ESG experience equips directors with valuable external resources, professional knowledge, and networks, thereby strengthening the board's oversight of environmental actions and fostering a culture of sustainability (Wang et al. 2025; Menicucci and Paolucci 2023). Consistent with our hypothesis H4, the results suggest that greater ESG experience enhances women directors' capacity to mitigate corporate environmental decoupling. In industries with low environmental sensitivity, ESG-experienced boards provide women directors with the tools and credibility needed to turn their stakeholder orientation and values into specific sustainability practices, ensuring that reporting reflects actual performance. Thus, ESG experience operates as a moderator that reinforces female directors' influence in reducing environmental decoupling.

6 Conclusions

This study explores the impact of board gender diversity on corporate environmental decoupling in firms within environmentally non-sensitive industries, emphasizing the moderating roles of board tenure and ESG experience. Our evidence highlights that women in the boardroom mitigate corporate environmental decoupling; nevertheless, this effect is contingent on various board characteristics. In this vein, the effect of female directors is accentuated in boards with longer tenure and greater ESG experience. These findings have important implications for businesses, financial markets, and society at large, and considering the ongoing debates around the role of board gender diversity in decision-making, particularly in

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First, our findings provide valuable insights for firms in environmentally non-sensitive industries. While these industries face less external pressure regarding environmental practices, current legislation and guidelines on environmental reporting affect them, making environmental decoupling potentially detrimental to their operations. As a result, firms must recognize the importance of properly addressing their environmental reporting practices. At the same time, companies are increasingly focusing on the composition of their boards. Thus, business cases that address board composition and environmental decoupling remain crucial for enhancing corporate accountability in this area. Our results can guide firms in the selection of board members, helping them understand the value added by key board attributes such as diversity, tenure, and ESG experience. From a strategic perspective, our study offers insights for nominating committees, stressing the need to consider the interactions among various board characteristics.

Second, our findings carry significant implications for professional organizations and policymakers, who have made recommendations regarding board gender diversity, board tenure, and board ESG experience. Our evidence offers a strong rationale for refining current legislation and recommendations to regulators and professionals in environmentally non-sensitive industries. It is particularly important to acknowledge that the decision-making behavior of female directors regarding environmental issues may be substantially influenced by other board characteristics. Moreover, our evidence provides valuable insights to regulatory authorities, highlighting the importance of both board tenure and board ESG experience, which are two factors that have been widely discussed, though explicit regulations on them remain lacking.

Third, from an academic standpoint, our study emphasizes the necessity of developing new theoretical frameworks to achieve a more comprehensive understanding of the role of board gender diversity. Our findings support the idea that female directors may help mitigate corporate environmental decoupling, yet it is crucial to extend theoretical research to explore the conditions that either amplify or reduce this effect. In this regard, our research contributes to the ongoing academic discourse by stressing the importance of moving beyond one-size-fits-all approaches.

Our study presents certain limitations that may be considered when designing future research. First, the sample variation in firm size is limited. To address this, future research could expand its scope to include small and medium-sized enterprises. Second, while we focus on a country that provides a relevant scenario for analyzing corporate governance

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Additionally, another promising avenue for future research could involve investigating the specific characteristics of female directors and their impact on corporate environmental decoupling or on other types of ESG decoupling. Finally, it is important to note that LSEG Eikon ESG scores are not available for all companies, which could limit the applicability of such studies.

Endnotes

¹ In line with the previous literature (Fernandez-Feijoo et al. [2014](#), Rudyanto and Veronica Siregar [2018](#)), environmentally non-sensitive industries are considered as those related to chemical, paper, metals, petroleum, mining and extractive, and utility industries- chemicals, metals, mining, paper, petroleum, or utilities industries, in accordance with SIC Codes classification).

² In the U.S. context, a board tenure of 10 years is considered optimal for ensuring effective board monitoring (Huang [2018](#)).

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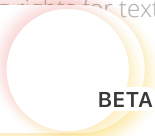
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